

Classification of quantum graphs over complex 3×3 matrices

(Klasyfikacja grafów kwantowych nad
zespolonymi macierzami 3×3)

Filip Rabięga

Master's thesis

Supervisor: Dr. Mariusz Tobolski

Uniwersytet Wrocławski
Wydział Matematyki i Informatyki
Instytut Matematyczny

February 31st, 2026

Abstract

According to all known laws of aviation, there is no way a bee should be able to fly. Its wings are too small to get its fat little body off the ground. The bee, of course, flies anyway because bees don't care what humans think is impossible.

Według wszystkich znanych praw lotnictwa, pszczoła nie powinna w ogóle latać. Jej skrzydła są zbyt małe, by unieść jej pulchne ciało z ziemi. Pszczoła, oczywiście, lata mimo wszystko, ponieważ pszczoły nie przejmują się tym, co ludzie uważają za niemożliwe.

Contents

1	Introduction	7
1.1	Motivation	7
1.2	Scope	7
1.3	Group theory	7
1.4	C^* -algebras	8
1.4.1	Crossed-product C^* -algebra	9
1.5	Quantum sets	9
1.6	Quantum graphs	10
2	Classification of quantum graphs over 3×3 matrices	11
2.1	Gromada's theorem	11

Chapter 1

Introduction

1.1 Motivation

1.2 Scope

1.3 Group theory

In the following section we will introduce some group theoretic notions that be used throughout this thesis. We assume that the Reader is familiar with the notion of a group.

Definition 1.3.1 (Dual of a group). Let G be a group. The *dual group* $\hat{G}$ is the set of all homomorphisms from G into the multiplicative group of roots of unity in $\mathbb{C}$ with the group operation defined as

$$(\varphi\psi)(g) = \varphi(g)\psi(g)$$

for all $\varphi, \psi \in \hat{G}$ and $g \in G$, where the latter multiplication takes place in $\mathbb{C}$. We will often call the members of the dual group *characters*.

Remark 1.3.1. We will often call the elements of the dual group *characters*. If a group is isomorphic to its complement it will be called *self-dual*.

The following proposition will be used throughout the thesis, as almost all of the groups we will be dealing with will be finite and abelian.

Proposition 1.3.1. *If G is finite and abelian, then it is self-dual.*

Proof. First we will prove the statement for finite cyclic groups $G = \langle g \rangle$, $\text{ord}(g) = n$. We will prove that the dual is cyclic of order n . Let $\chi : G \rightarrow \mathbb{C}$ be a character. Since the group is cyclic, it is entirely determined by the value it takes on its generator g . Since $g^n = e$, obviously $\chi(g)^n = 1$, so $\chi(g)^n \in \mu_n$. For each possible choice of

$\chi(g) \in \mu_n$ we get a different character. It is then easy to see that $\chi : G \rightarrow \mathbb{C}$ such that $\chi(g) = \exp(2\pi i/n)$ generates this group.

Now since every finite abelian group can be written as a finite product of finite cycles, all we need to do is check whether the statement holds for a product of two groups A, B that are themselves self-dual, i.e. $\widehat{A \times B} \cong \widehat{A} \times \widehat{B}$. This is self-evident as every character $\chi \in \widehat{A \times B}$ is determined by its restrictions to $A \times \{e_B\}$ and $\{e_A\} \times B$, since

$$\chi(a, b) = \chi((a, e)(e, b)) = \chi(a, e)\chi(e, b).$$

Thus $\chi \leftrightarrow (\chi|_A, \chi|_B)$ is an isomorphism. $\square$

Definition 1.3.2 (Group action on a C^* -algebra). By *group action* of a group G on a C^* -algebra A we mean a homomorphism

$$\alpha : G \rightarrow \text{Aut}(A).$$

We will write $g \cdot a$ instead of $\alpha(g)(a)$ then the action in question is clear.

1.4 C^* -algebras

In the following section we will introduce some notions stemming from the theory of C^* -algebras.

Definition 1.4.1 (Linear functional). A *linear functional* on a C^* -algebra A is a linear function $\psi : A \rightarrow \mathbb{C}$.

Remark 1.4.1. We will call linear functionals simply as *functionals*.

The following notions will be useful for defining an inner product on C^* -algebras. Positiveness and faithfulness is all we need to define a norm on a C^* -algebra equipped with a suitable functional.

Definition 1.4.2 (Positive functional). A functional ψ on a C^* -algebra A is called *positive*, if

$$\psi(a^*a) \geq 0$$

for any $a \in A$.

Definition 1.4.3 (Faithful functional). A positive functional ψ on a C^* -algebra A is called *faithful*, if

$$\psi(a^*a) = 0 \Leftrightarrow a = 0$$

for any $a \in A$.

1.4.1 Crossed-product C^* -algebra

In this section we will introduce the notion of a crossed-product C^* -algebra. We will do so in a limited manner, only to the extent to which the theory of such products is necessary for our purposes. For a more general reference we refer the Reader to Blackadar [1].

Definition 1.4.4 (C^* -dynamical system). A C^* -dynamical system is a triple (A, G, α) , where A is a C^* -algebra, G is a group, and $\alpha : G \rightarrow \text{Aut}(A)$ is a group action on A . For a C^* dynamical system (A, G, α) with G finite we define

$$C_c(G, A, \alpha) = \{f : G \rightarrow A\},$$

i.e. the space of all functions on G with values in A .

$C_c(G, A, \alpha)$ is equipped with both a product and a $*$ -operation in such a way that the action becomes "inner", i.e. $g \cdot a = u_g a u_g^*$ for all $g \in G$, $a \in A$. More precisely we define those two operations in the following way:

$$ab = \sum_{g, h \in G}$$

Definition 1.4.5 (Cross-product C^* -algebra). Let A be a C^* -algebra and let G be a group acting on A . blabl

1.5 Quantum sets

Let B denote a C^* -algebra and let ψ denote a functional on this algebra. By the Gelfand-Naimark theorem, every C^* -algebra is isometrically $*$ -isomorphic to a C^* -subalgebra $B(\mathcal{H})$ for some Hilbert space $\mathcal{H}$. If B is finite-dimensional, one can identify $L^2(B)$ with B and think of B as a Hilbert space with an inner product given by

$$\langle a, b \rangle_\psi = \psi(a^* b).$$

The following definition introduces one of the central notions used in this thesis. It is taken directly from Schäfer i Tobolski [3].

Definition 1.5.1 (Quantum set, Schäfer i Tobolski [3], Definition 2.1). A quantum set (B, ψ) is a pair consisting of a finite-dimensional C^* -algebra B and a faithful positive functional $\psi : B \rightarrow \mathbb{C}$ such that

$$mm^* = \text{Id}_B$$

holds for the multiplication map $m : B \otimes B \rightarrow B$ and its adjoint m^* given by

$$\langle m(a \otimes b), c \rangle = \langle a \otimes b, m^*(c) \rangle_{\psi \otimes \psi}$$

for all $a, b, c \in B$.

Remark 1.5.1. We will call any functional with the property given in the definition above a *quantum set functional* on B .

1.6 Quantum graphs

Chapter 2

Classification of quantum graphs over 3×3 matrices

2.1 Gromada's theorem

In the following section we will state...

The following theorem is due to Gromada [2].

Theorem 2.1.1 (Gromada [2], Proposition 3.1). *Let A be a finite quantum set. A quantum graph...*

Bibliografia

- [1] B. Blackadar. *Operator Algebras: Theory of C^* -Algebras and von Neumann Algebras*. Encyclopaedia of Mathematical Sciences. Springer Berlin Heidelberg, 2006. ISBN: 9783540285175. URL: <https://books.google.pl/books?id=7-6MZLdRfdAC>.
- [2] Daniel Gromada. “Some examples of quantum graphs”. W: *Letters in Mathematical Physics* 112.6 (list. 2022). ISSN: 1573-0530. DOI: 10.1007/s11005-022-01603-5. URL: <http://dx.doi.org/10.1007/s11005-022-01603-5>.
- [3] Björn Schäfer i Mariusz Tobolski. “Voltage Quantum Graphs and a Gross–Tucker Theorem for Quantum Graphs”. W: (2026). arXiv: 2602.20996 [math.OA].